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Agniveer

Weekly Study Digest

June 1 – June 7, 2026

15 Questions . Detailed Solutions with Shortcuts

Topic Analysis

1. Topics Covered - Mathematics, Science, History, Geography, and General Knowledge
2. Difficulty Breakdown - 40% easy, 30% medium, 30% hard
3. Key Concepts - Basic math operations, scientific terminology, historical events, and geographical facts
4. Common Mistakes - Calculation errors, incorrect units, and lack of basic knowledge
5. Study Tips - Practice math problems, learn scientific terms, and memorize historical and geographical facts

& Quick Reference

6. Key Formulas & Shortcuts - * Basic algebra formulas, * unit conversions, * and mathematical shortcuts
7. Weekly Strategy - * Focus on one subject per day, * practice mixed-question sets, and * review notes regularly

Reasoning

1 If a person has 12 pencils and gives $\frac{1}{4}$ of them to their friend, how many pencils does the person have left?

- ☒ A. 9
☐ B. 10
☐ C. 11
☐ D. 12

Correct: A

Detailed Solution

To solve this problem, we need to find out how many pencils the person gives to their friend and then subtract that number from the total number of pencils they initially had. The person has 12 pencils and gives $\frac{1}{4}$ of them to their friend. To find $\frac{1}{4}$ of 12, we can multiply 12 by $\frac{1}{4}$, which is the same as dividing 12 by 4. So, $12 \times \frac{1}{4} = 12 / 4 = 3$. This means the person gives 3 pencils to their friend. Now, we subtract 3 from the total number of pencils: $12 - 3 = 9$. Therefore, the person has 9 pencils left. A common mistake students make is to forget to subtract the number of pencils given away from the total. A smart shortcut to solve this problem is to remember that $\frac{1}{4}$ of a number is the same as dividing that number by 4. The logical reasoning behind this step is to understand that when we give away a fraction of something, we need to find out how much that fraction is worth and then subtract it from the total.

The final answer is A.

Science

2

What is the scientific term for the 'building blocks of life'?

- ☒ A. Cells
- ☐ B. Molecules
- ☐ C. Tissues
- ☐ D. Organisms

' Correct: A

Detailed Solution

The scientific term for the 'building blocks of life' is Cells. This is because cells are the basic structural and functional units of living organisms. All living things are composed of one or more cells, and cells are the smallest units of life that can replicate independently. The other options, Molecules, Tissues, and Organisms, are not the building blocks of life. Molecules are the basic units of a chemical compound, tissues are groups of cells that work together, and organisms are individual living things. A common mistake students make is to confuse cells with other levels of organization in living things. A smart shortcut to remember this is to recall that cells are the basic units of life and that all living things are made up of cells. The logical reasoning behind this is to understand the definition of cells and their role in living organisms.

The final answer is A.

General Knowledge

3

Who was the founder of the Maurya Empire?

- ☐ A. Bindusara
- ☐ B. Ashoka
- ☒ C. Chandragupta Maurya
- ☐ D. Samudragupta

' Correct: C

Detailed Solution

The founder of the Maurya Empire was Chandragupta Maurya. He was an Indian emperor who ruled from 322 BCE to 298 BCE and is considered one of the most important figures in Indian history. Chandragupta Maurya was the first emperor to unify India under a single rule, and his empire stretched from present-day Afghanistan to Bangladesh. The other options, Bindusara, Ashoka, and Samudragupta, were also important figures in Indian history, but they were not the founders of the Maurya Empire. Bindusara was the son of Chandragupta Maurya, Ashoka was the grandson of Chandragupta Maurya, and Samudragupta was a Gupta emperor who ruled much later. A common mistake students make is to confuse the different emperors and their roles in Indian history. A smart shortcut to remember this is to recall that Chandragupta Maurya was the first emperor to unify India. The logical reasoning behind this is to understand the timeline of Indian history and the roles of different emperors.

The final answer is C.

Science

4

What is the process by which water moves through a plant, from the roots to the leaves?

- ☐ A. Respiration
- ☐ B. Photosynthesis
- ☒ C. Transpiration
- ☐ D. Evaporation

Correct: C

Detailed Solution

The process by which water moves through a plant, from the roots to the leaves, is called Transpiration. This process occurs through the xylem, a type of vascular tissue in plants, and is an important part of the water cycle. Transpiration is the process by which plants release water vapor into the air, and it helps to regulate the plant's temperature and maintain its water balance. The other options, Respiration, Photosynthesis, and Evaporation, are all important processes in plants, but they are not the same as transpiration. Respiration is the process by which plants break down glucose to release energy, photosynthesis is the process by which plants produce glucose from sunlight, and evaporation is the process by which water changes from a liquid to a gas. A common mistake students make is to confuse these different processes. A smart shortcut to remember this is to recall that transpiration is the process by which plants release water vapor into the air. The logical reasoning behind this is to understand the different processes that occur in plants and their roles in the water cycle.

The final answer is C.

Mathematics

5

What is the value of x in the equation $2x + 2 = 12$?

- ☐ A. 4
- ☒ B. 5
- ☐ C. 6
- ☐ D. 8

Correct: B

Detailed Solution

To solve this equation, we need to isolate the variable x . The equation is $2x + 2 = 12$. To isolate x , we first subtract 2 from both sides of the equation: $2x + 2 - 2 = 12 - 2$. This simplifies to $2x = 10$. Next, we divide both sides of the equation by 2: $2x / 2 = 10 / 2$. This simplifies to $x = 5$. Therefore, the value of x is 5. A common mistake students make is to forget to perform the same operation on both sides of the equation. A smart shortcut to solve this problem is to remember to isolate the variable by performing inverse operations. The logical reasoning behind this is to understand that the goal of solving an equation is to isolate the variable, and that we can do this by performing inverse operations.

The final answer is B.

Reasoning

6

If it takes 5 machines 5 minutes to make 5 widgets, how long would it take 100 machines to make 100 widgets?

- ☒ A. 5 minutes
- ☐ B. 10 minutes
- ☐ C. 50 minutes
- ☐ D. 100 minutes

' Correct: A

Detailed Solution

To solve this problem, we need to understand the relationship between the number of machines and the time it takes to make widgets. If it takes 5 machines 5 minutes to make 5 widgets, then it means that each machine takes 5 minutes to make 1 widget. If we have 100 machines, it will still take each machine 5 minutes to make 1 widget. Therefore, it will take 100 machines 5 minutes to make 100 widgets. A common mistake students make is to think that the time it takes to make widgets will increase as the number of machines increases. A smart shortcut to remember this is to recall that the time it takes to make widgets is determined by the rate at which each machine works, not by the number of machines. The logical reasoning behind this is to understand the relationship between the number of machines and the time it takes to make widgets, and to recognize that the rate at which each machine works is the key factor.

The final answer is A.

General Knowledge

7

Which river is the longest in India?

- ☒ A. Ganga
- ☐ B. Brahmaputra
- ☐ C. Yamuna
- ☐ D. Godavari

' Correct: A

Detailed Solution

The longest river in India is the Ganga. The Ganga River is approximately 2,525 kilometers long and flows through several states in northern India, including Uttarakhand, Uttar Pradesh, Bihar, and West Bengal. The other options, Brahmaputra, Yamuna, and Godavari, are all important rivers in India, but they are not the longest. The Brahmaputra River is approximately 2,900 kilometers long, but most of its length is outside of India. The Yamuna River is a tributary of the Ganga River, and the Godavari River is a major river in southern India. A common mistake students make is to confuse the different rivers in India and their lengths. A smart shortcut to remember this is to recall that the Ganga River is the longest river in India. The logical reasoning behind this is to understand the geography of India and the lengths of its major rivers.

The final answer is A.

General Knowledge

8

Which country is the largest producer of milk in the world?

- ☒ A. India
- ☐ B. United States
- ☐ C. China
- ☐ D. Brazil

' Correct: A

Detailed Solution

The largest producer of milk in the world is India. India produces over 180 million tons of milk per year, accounting for over 20% of global milk production. The other options, United States, China, and Brazil, are all significant milk producers, but they do not produce as much milk as India. The United States is the second-largest milk producer, followed by China and Brazil. A common mistake students make is to think that the United States or China is the largest milk producer. A smart shortcut to remember this is to recall that India has a large dairy industry and is the largest producer of milk in the world. The logical reasoning behind this is to understand the global dairy industry and the major milk-producing countries.

The final answer is A.

Reasoning

9

A person has a set of 5 keys, and only one of them opens a certain door. If the person tries the keys one by one, what is the probability that the person will open the door on the third attempt?

- ☒ A. 1/5
- ☐ B. 1/3
- ☐ C. 2/5
- ☐ D. 3/5

' Correct: A

Detailed Solution

To solve this problem, we need to understand the concept of probability. The person has a set of 5 keys, and only one of them opens the door. If the person tries the keys one by one, the probability of opening the door on the first attempt is $1/5$, on the second attempt is $1/4$ (since one key has already been tried), and on the third attempt is $1/3$ (since two keys have already been tried). However, since the person is trying the keys one by one, the probability of opening the door on the third attempt is actually the probability of not opening the door on the first two attempts and then opening it on the third attempt. This is equal to $(4/5) * (3/4) * (1/3) = 1/5$. A common mistake students make is to confuse the different probabilities and to think that the probability of opening the door on the third attempt is $1/3$. A smart shortcut to remember this is to recall that the probability of an event is equal to the number of favorable outcomes divided by the total number of outcomes. The logical reasoning behind this is to understand the concept of probability and to apply it to the given situation.

The final answer is A.

Mathematics

10 What is the perimeter of a rectangle with a length of 6 cm and a width of 4 cm?

- ☐ A. 16 cm
☐ B. 18 cm
☒ C. 20 cm
☐ D. 22 cm

' Correct: C

Detailed Solution

To find the perimeter of a rectangle, we need to add up the lengths of all its sides. The formula for the perimeter of a rectangle is $P = 2(l + w)$, where l is the length and w is the width. In this case, the length is 6 cm and the width is 4 cm. Plugging these values into the formula, we get $P = 2(6 + 4) = 2(10) = 20$ cm. Therefore, the perimeter of the rectangle is 20 cm. A common mistake students make is to forget to multiply the sum of the length and width by 2. A smart shortcut to remember this is to recall that the perimeter of a rectangle is equal to twice the sum of its length and width. The logical reasoning behind this is to understand the formula for the perimeter of a rectangle and to apply it to the given situation.

The final answer is C.

Mathematics

11 What is the perimeter of a square with side length 4 cm?

- ☐ A. 12 cm
☐ B. 14 cm
☒ C. 16 cm
☐ D. 20 cm

' Correct: C

Detailed Solution

To find the perimeter of a square, we need to add up the lengths of all its sides. Since all sides of a square are equal, the formula for the perimeter of a square is $P = 4s$, where s is the length of one side. In this case, the side length is 4 cm. Plugging this value into the formula, we get $P = 4(4) = 16$ cm. Therefore, the perimeter of the square is 16 cm. A common mistake students make is to forget that all sides of a square are equal. A smart shortcut to remember this is to recall that the perimeter of a square is equal to four times the length of one side. The logical reasoning behind this is to understand the properties of a square and to apply the formula for its perimeter.

The final answer is C.

Reasoning

- 12 If a snail is at the bottom of a 20-foot well, and it climbs 3 feet up the well each day, but slips back 2 feet at night, how many days will it take for the snail to reach the top of the well?

- ☒ A. 18 days
☐ B. 20 days
☐ C. 22 days
☐ D. 25 days

' Correct: A

Detailed Solution

To solve this problem, we need to understand the concept of net progress. The snail climbs 3 feet up the well each day, but slips back 2 feet at night, so its net progress is $3 - 2 = 1$ foot per day. Since the well is 20 feet deep, the snail will need 20 days to reach the top, but on the 18th day, it will climb 3 feet and reach the top of the well, so it will not slip back on the 18th night. Therefore, it will take the snail 18 days to reach the top of the well. A common mistake students make is to think that the snail will need 20 days to reach the top. A smart shortcut to remember this is to recall that the snail's net progress is 1 foot per day, and to calculate the number of days it will take to reach the top. The logical reasoning behind this is to understand the concept of net progress and to apply it to the given situation.

The final answer is A.

Reasoning

- 13 If it takes 3 hours to walk 9 miles, how many hours will it take to walk 12 miles at the same rate?

- ☐ A. 3 hours
☒ B. 4 hours
☐ C. 5 hours
☐ D. 6 hours

' Correct: B

Detailed Solution

To solve this problem, we need to understand the concept of rate and time. If it takes 3 hours to walk 9 miles, then the rate at which the person walks is $9 \text{ miles} / 3 \text{ hours} = 3 \text{ miles per hour}$. To find the time it will take to walk 12 miles at the same rate, we can divide the distance by the rate: $12 \text{ miles} / 3 \text{ miles per hour} = 4 \text{ hours}$. Therefore, it will take 4 hours to walk 12 miles at the same rate. A common mistake students make is to think that the time it takes to walk a certain distance is directly proportional to the distance, without considering the rate. A smart shortcut to remember this is to recall that the time it takes to travel a certain distance is equal to the distance divided by the rate. The logical reasoning behind this is to understand the concept of rate and time and to apply it to the given situation.

The final answer is B.

General Knowledge

14 Which article of the Indian Constitution grants the 'Right to Equality'?

- ☐ A. Article 19
- ☒ B. Article 14
- ☐ C. Article 21
- ☐ D. Article 32

' Correct: B

Detailed Solution

The article of the Indian Constitution that grants the 'Right to Equality' is Article 14. Article 14 states that the state shall not deny to any person equality before the law or the equal protection of the laws within the territory of India. This article is a fundamental right that is guaranteed to all citizens of India, and it is an important part of the Indian Constitution. The other options, Article 19, Article 21, and Article 32, are all important articles in the Indian Constitution, but they do not grant the right to equality. Article 19 grants the right to freedom, Article 21 grants the right to life and personal liberty, and Article 32 grants the right to constitutional remedies. A common mistake students make is to confuse the different articles in the Indian Constitution and their provisions. A smart shortcut to remember this is to recall that Article 14 grants the right to equality. The logical reasoning behind this is to understand the provisions of the Indian Constitution and to apply them to the given situation.

The final answer is B.

General Knowledge

15 Which Indian state has the highest number of UNESCO World Heritage Sites as of 2024?

- ☐ A. Kerala
- ☐ B. Tamil Nadu
- ☐ C. Madhya Pradesh
- ☒ D. Uttar Pradesh

' Correct: D

Detailed Solution

The Indian state with the highest number of UNESCO World Heritage Sites as of 2024 is Uttar Pradesh. Uttar Pradesh has a total of 5 UNESCO World Heritage Sites, including the Taj Mahal, the Agra Fort, and the Fatehpur Sikri. The other options, Kerala, Tamil Nadu, and Madhya Pradesh, are all important states in India with a rich cultural heritage, but they do not have as many UNESCO World Heritage Sites as Uttar Pradesh. Kerala has 2 UNESCO World Heritage Sites, Tamil Nadu has 4, and Madhya Pradesh has 3. A common mistake students make is to think that the state with the most UNESCO World Heritage Sites is Tamil Nadu or Madhya Pradesh. A smart shortcut to remember this is to recall that Uttar Pradesh has the most UNESCO World Heritage Sites in India. The logical reasoning behind this is to understand the cultural heritage of India and to apply it to the given situation.

The final answer is D.

Study Tips

Maximize your preparation with these strategies:

[R] Review Thoroughly

Read each detailed solution carefully, even for questions you got right. The shortcuts and alternative methods will save you time on exam day.

[P] Practice Similar Problems

After reviewing, solve 2-3 similar problems on the same topic. Pattern recognition is the key to speed in competitive exams.

[S] Master the Shortcuts

The shortcuts in these solutions are battle-tested. Practice them until they become automatic. A 30-second shortcut can decide your rank.

[T] Track Accuracy & Speed

Note time per question. If a question takes more than 2 minutes, flag it. Review why it took long - was it the concept or the calculation?

[F] Focus on Weak Areas

Use the Topic Analysis to identify weak spots. Dedicate extra sessions to those topics before moving to new material.

[E] Simulate Exam Conditions

Take timed mock tests on vlymbooq. Real exam simulation reduces anxiety and builds the mental stamina needed for 2-3 hour papers.